Improving CLIP Training: A Comparative Study of Loss Functions and Optimizers Under Resource Constraints

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Abstract

Contrastive Language-Image Pre-training (CLIP) has revolutionized vision-language understanding, yet its training efficiency remains constrained by severe batch-size sensitivity and substantial computational requirements. This work presents a empirical investigation of alternative loss functions and optimizer combinations to improve CLIP training at reduced computational scales. We evaluate eight loss functions, including standard InfoNCE (CLIP), global contrastive objectives (SogCLR, iSogCLR), geometric constraints (CyCLIP), pairwise losses (SigLIP), non-contrastive methods (VICReg), and two novel hybrid approaches (SigLIP+CyCLIP, iSogCLR+CyCLIP), across seven modern optimizers (AdamW, AdamP, RAdam, LAMB, Lion, SGDP, Adafactor) on a 100k subset of CC3M. Our comprehensive experiments, utilizing nearly 25,000 service units on TAMU's Grace/ACES clusters, reveal that global contrastive learning with adaptive per-sample temperature scaling (iSogCLR) paired with adaptive gradient clipping (AdamP) achieves optimal performance, reaching 27.6% zero-shot accuracy@1 on ImageNet and 18.5% average (zero-shot accuracy, text recall and image recall)@1 across tasks with only 128 batch size. We demonstrate that careful algorithmic design can reduce batch-size requirements by 250× (from 32k to 128) while maintaining competitive performance, making vision-language pre-training accessible to resource-constrained research environments.

1 Introduction

Self-supervised learning has fundamentally transformed representation learning across computer vision and natural language processing, enabling models to learn rich semantic representations from vast amounts of unlabeled data. Among these approaches, Contrastive Language-Image Pre-training (CLIP) (1) has demonstrated remarkable zero-shot transfer capabilities by learning aligned multi-modal representations through contrastive learning on 400 million image-text pairs harvested from the internet.

Despite its transformative impact, CLIP's training methodology presents several critical practical challenges that limit its accessibility and applicability. **First**, CLIP exhibits severe batch-size sensitivity, typically requiring massive batch sizes (32,768 samples) to achieve competitive performance. making the contrastive task less informative and resulting in poor generalization. **Second**, the computational requirements are extremely high, demanding hundreds of high-end GPUs (e.g., 256 V100s) and weeks of continuous training, placing CLIP beyond reach for most academic research groups and small organizations. **Third**, CLIP often converges slowly because its contrastive loss must balance alignment and uniformity over very large batches, making optimization sensitive and requiring many training steps. **Fourth**, CLIP uses a single global temperature that cannot adapt to the varying difficulty of image–text pairs, which several works show leads to suboptimal gradients and reduced training stability.

Recent work has begun addressing these limitations through algorithmic innovations. **Global contrastive learning** approaches like SogCLR (2) and iSogCLR (3) approximate global objectives using moving averages, reducing batch-size sensitivity through provably convergent stochastic optimization. **Geometric consistency** methods like CyCLIP (4) enforce structural constraints on the embedding space, improving alignment without requiring larger batches. **Pairwise formulations** like SigLIP (5) replace batch-level normalization with independent pair-wise predictions, achieving natural batch-size invariance. However, systematic comparisons of these methods under controlled conditions remain limited, and the interaction between loss functions and optimizers is under-explored.

This work addresses these gaps through a comprehensive empirical study of loss functions and optimizers for vision-language pre-training at reduced computational scales. Our contributions are:

- **Comprehensive loss function comparison**: We systematically evaluate eight loss functions spanning contrastive (CLIP, SogCLR, iSogCLR), geometric (CyCLIP), pairwise (SigLIP), and non-contrastive (VICReg) paradigms, revealing their relative strengths under resource constraints.

- **Novel hybrid approaches**: We propose and evaluate two hybrid loss combinations—SigLIP+CyCLIP and iSogCLR+CyCLIP—demonstrating how complementary objectives can be combined, though revealing trade-offs between zero-shot classification and retrieval performance.

- **Optimizer sensitivity analysis**: We evaluate seven modern optimizers (AdamW, AdamP, RAdam, LAMB, Lion, SGDP, Adafactor) across all loss functions, revealing critical interactions between optimization algorithms and loss landscapes, with AdamP emerging as the most robust choice.

2 Related Work

2.1 Contrastive Language-Image Learning

CLIP (1) pioneered large-scale vision-language pre-training using InfoNCE loss on 400M image-text pairs with batch sizes of 32,768. Despite remarkable zero-shot capabilities, CLIP's severe batch-size sensitivity and computational requirements (256 V100 GPUs for two weeks) motivated research into more efficient alternatives. **OpenCLIP** (7) confirmed that reducing batch size from 32k to 4k causes significant performance degradation, highlighting the need for algorithmic solutions.

2.2 Global Contrastive Learning

SogCLR (2) addresses batch-size limitations by reformulating the objective as a global contrastive loss contrasting each sample against all dataset samples, not just batch samples. The method uses exponential moving averages to approximate global objectives, with provable convergence guarantees showing optimization error depends on dataset size rather than batch size.

iSogCLR (3) extends SogCLR with automatic per-sample temperature individualization through Distributionally Robust Optimization (DRO). Hard samples receive lower temperatures (sharper distributions) to focus learning, while easy samples receive higher temperatures to prevent overfitting. This adaptive mechanism eliminates manual temperature tuning and improves sample efficiency.

TempNet (20) further advances temperature learning by parameterizing temperatures as neural network outputs rather than per-sample scalars. This approach enables temperature generalization to unseen samples during inference and provides more flexible temperature adaptation through learned feature-dependent functions. The method demonstrates improved zero-shot transfer by learning how sample difficulty correlates with visual and textual features.

FastCLIP (8) recently scaled these methods to 315M image-text pairs, demonstrating that global contrastive approaches can match OpenCLIP's performance with 6× smaller batch sizes (5,120 vs 33,792).

2.3 Geometric Consistency and Alternative Formulations

CyCLIP (4) enforces cycle consistency constraints ensuring: (1) image-to-image similarities match text-to-text similarities (intra-modal consistency), and (2) image-to-text similarities equal text-to-image similarities (cross-modal consistency). These geometric regularizers improve embedding quality without requiring larger batches.

SigLIP (5) replaces InfoNCE's batch-level softmax with independent pairwise sigmoid losses, treating image-text matching as B^2 binary classification problems. This formulation provides natural batch-size invariance since pairs are predicted independently, with a learnable bias controlling the decision boundary.

VICReg (6) proposes non-contrastive learning using variance-invariance-covariance regularization. Originally designed for unimodal SSL, we adapt VICReg to vision-language learning by treating image and text as "two views," testing whether non-contrastive objectives suffice for cross-modal alignment.

2.4 Optimizer Innovations

Modern optimizers address large-scale training challenges through various mechanisms. **AdamP** (9) uses projection-based gradient clipping for stability in sharp loss regions. **RAdam** (10) rectifies adaptive learning rates during warm-up when momentum estimates lack statistics. **LAMB** (11) enables layer-wise adaptation for extremely large batches. **Lion** (12) uses sign-based updates with momentum, requiring less memory but extreme sensitivity to learning rates. Our work systematically evaluates these optimizers for vision-language contrastive learning, revealing strong loss-optimizer interactions.

3 Method

3.1 Problem Formulation

We consider a dataset $\mathcal{D} = \{(x_i, t_i)\}_{i=1}^N$ of N image-text pairs. Our goal is to learn visual encoder $f_v : \mathcal{X} \to \mathbb{R}^d$ and text encoder $f_t : \mathcal{T} \to \mathbb{R}^d$ that map to a shared embedding space where semantically related pairs have high cosine similarity. Let $v_i = f_v(x_i)$ and $e_i = f_t(t_i)$ denote ℓ_2-normalized embeddings with similarity $s_{ij} = v_i^\top e_j$.

3.2 Loss Functions

3.2.1 CLIP: InfoNCE Contrastive Loss

CLIP uses symmetric InfoNCE loss treating image-text matching as multi-class classification over batch $\mathcal{B}$:

$$\mathcal{L}_{\text{CLIP}} = -\frac{1}{2B} \sum_{i \in \mathcal{B}} \left[\log \frac{\exp(s_{ii}/\tau)}{\sum_j \exp(s_{ij}/\tau)} + \log \frac{\exp(s_{ii}/\tau)}{\sum_j \exp(s_{ji}/\tau)} \right] \tag{1}$$

where τ is a learnable temperature. Performance critically depends on batch size B since negatives are drawn exclusively from the current batch.

3.2.2 SogCLR: Global Contrastive Loss

SogCLR approximates a global contrastive loss over all dataset samples using exponential moving averages. For each sample i, maintain:

$$u_{1,i,t} = (1-\gamma) u_{1,i,t-1} + \gamma \cdot \frac{1}{|\mathcal{B}_i^-|} \sum_{j \in \mathcal{B}_i^-} \exp\left(\frac{s_{ij} - s_{ii}}{\tau}\right) \tag{2}$$

where $\mathcal{B}_i^- = \mathcal{B} \setminus \{i\}$ and $\gamma \in (0, 1]$ is the moving average rate. The gradient estimator becomes:

$$\nabla_w \mathcal{L}_{\text{SogCLR}} \approx \frac{\tau}{B} \sum_{i \in \mathcal{B}} \frac{1}{\epsilon + u_{1,i,t}} \cdot \nabla_w g_1(w, \tau, i, \mathcal{B}_i^-) \tag{3}$$

This provides global information without $O(N^2)$ computation, reducing batch-size sensitivity.

3.2.3 iSogCLR: Adaptive Temperature Individualization

iSogCLR extends SogCLR with per-sample learnable temperatures $\{\tau_{1,i}, \tau_{2,i}\}$ via Distributionally Robust Optimization:

$$\min_{w} \max_{\tau \geq \tau_0} \frac{1}{N} \sum_{i=1}^{N} \left[\tau_{1,i} \left(\log(\epsilon + g_1) + \rho\right) + \tau_{2,i} \left(\log(\epsilon + g_2) + \rho\right)\right] \tag{4}$$

where $\rho > 0$ regularizes temperature magnitude. Temperatures are updated via gradient ascent, with hard samples receiving lower temperatures (sharper distributions) and easy samples higher temperatures (softer distributions). Constraints $\tau_{\min} \leq \tau_i \leq \tau_{\max}$ prevent collapse.

3.2.4 CyCLIP: Cycle Consistency

CyCLIP adds geometric regularization to CLIP:

$$\mathcal{L}_{\text{CyCLIP}} = \mathcal{L}_{\text{CLIP}} + \lambda_2 \mathcal{L}_{\text{intra}} + \lambda_3 \mathcal{L}_{\text{cross}} \tag{5}$$

where intra-modal consistency enforces $S_{II} \approx S_{TT}$ (image-image similarities match text-text similarities):

$$\mathcal{L}_{\text{intra}} = \frac{1}{B^2} \|S_{II} - S_{TT}\|_F^2 \tag{6}$$

and cross-modal consistency enforces $S \approx S^\top$ (bidirectional alignment):

$$\mathcal{L}_{\text{cross}} = \frac{1}{B^2} \left\|S - S^\top\right\|_F^2 \tag{7}$$

We use $\lambda_2 = \lambda_3 = 0.25$ following (4).

3.2.5 SigLIP: Sigmoid Pairwise Loss

SigLIP replaces softmax with independent sigmoid losses:

$$\mathcal{L}_{\text{SigLIP}} = -\frac{1}{B^2} \sum_{i,j \in \mathcal{B}} \left[y_{ij} \log \sigma(z_{ij}) + (1 - y_{ij}) \log(1 - \sigma(z_{ij}))\right] \tag{8}$$

where $y_{ij} = \mathbb{1}[i = j]$ and $z_{ij} = s_{ij}/\tau + b$ with learnable bias b (initialized to -10). This formulation provides natural batch-size invariance since pairs are predicted independently.

3.2.6 VICReg: Non-Contrastive Regularization

We adapt VICReg to bimodal learning by treating image and text as "two views":

$$\mathcal{L}_{\text{VICReg}} = \lambda_{\text{inv}} \mathcal{L}_{\text{inv}} + \lambda_{\text{var}} \mathcal{L}_{\text{var}} + \lambda_{\text{cov}} \mathcal{L}_{\text{cov}} \tag{9}$$

with invariance (MSE between embeddings), variance (prevent collapse), and covariance (decorrelate dimensions) terms. We use $(\lambda_{\text{inv}}, \lambda_{\text{var}}, \lambda_{\text{cov}}) = (25, 25, 1)$.

3.3 Novel Hybrid Approaches

3.3.1 SigLIP + CyCLIP

Combines SigLIP's pairwise independence with CyCLIP's geometric constraints:

$$\mathcal{L}_{\text{SigLIP+CyCLIP}} = \mathcal{L}_{\text{SigLIP}} + \lambda_2 \mathcal{L}_{\text{intra}}^{\text{sig}} + \lambda_3 \mathcal{L}_{\text{cross}}^{\text{sig}} \tag{10}$$

Cycle terms are computed in sigmoid space for consistency, with $\mathcal{L}_{\text{intra}}^{\text{sig}} = \frac{1}{B^2} \sum_{i,j} [\sigma(z_{II,ij}) - \sigma(z_{TT,ij})]^2$.

3.3.2 iSogCLR + CyCLIP

Combines iSogCLR's adaptive temperatures with CyCLIP's structure:

$$\mathcal{L}_{\text{iSogCLR+CyCLIP}} = \mathcal{L}_{\text{iSogCLR}} + \lambda_2 \mathcal{L}_{\text{intra}}^{\text{adap}} + \lambda_3 \mathcal{L}_{\text{cross}}^{\text{adap}} \tag{11}$$

Since iSogCLR has per-sample temperatures, we use average temperature $\tau_{\text{avg}} = \frac{1}{2B} \left(\sum_i \tau_{1,i} + \sum_i \tau_{2,i} \right)$ for cycle computations.

3.4 Optimizer Selection

We evaluate seven optimizers: **AdamW** (13) (decoupled weight decay), **AdamP** (9) (projection-based gradient clipping), **RAdam** (10) (rectified adaptive learning rates), **LAMB** (11) (layer-wise adaptation), **Lion** (12) (sign-based updates), **SGDP** (9) (SGD with projection), and **Adafactor** (14) (memory-efficient factorized moments). Detailed hyperparameters are provided in Section 4.4.

4 Experimental Results

4.1 Dataset and Architecture

We train on a 100k subset of Conceptual Captions 3M (CC3M) (15) for 30 epochs with batch size 128. Models use pretrained ResNet-50 (18) image encoder and DistilBERT (19) text encoder with frozen architectures.

Evaluation uses:

- **MSCOCO retrieval** (16): Text-to-image (5,000 images) and image-to-text (25,000 captions). Report Recall@1, @5, @10.
- **ImageNet zero-shot classification** (17): 1,000-class classification. Report Top-1, Top-3, Top-5, Top-10 accuracy.
- **Primary metric**: Average Recall@1 = mean of (ZS@1, TR@1, IR@1).

4.2 Hyperparameters

We adapted hyperparameters from CLIP literature (1; 2; 3; 12) to our training scale (100k samples, 30 epochs, batch 128). Table 1 shows selected values. Other settings: warmup=20 epochs, temperature $\tau = 0.01$, iSogCLR parameters $\gamma = 0.8$, $\eta = 0.03$, $\rho = 8.0$.

4.3 Top-10 Overall Results

Table 2 shows the top-10 configurations by Average Recall@1. iSogCLR with adaptive optimizers (AdamP, RAdam, LAMB) dominates the leaderboard.

4.4 Best Performance Per Loss Function

Table 3 shows the best optimizer for each loss function. Global contrastive methods (iSogCLR, SogCLR) significantly outperform batch-based methods (CLIP, SigLIP).

Table 1: Optimizer hyperparameters adapted from literature for our training scale.

Optimizer	Learning Rate	Betas / Momentum	Weight Decay
AdamW	1e-3	(β_1=0.9, β_2=0.999), eps=1e-8	0.01
AdamP	1e-4	(β_1=0.9, β_2=0.98)	0.1
RAdam	1e-4	(β_1=0.9, β_2=0.98)	0.1
Lion	3e-5	(β_1=0.9, β_2=0.99)	0.5
LAMB	2e-3	(β_1=0.9, β_2=0.999)	0.01
SGDP	0.1	momentum=0.9, nesterov=True	1e-4
Adafactor	1e-3	(adaptive)	0.01

Table 2: Top-10 configurations by Average Recall@1 (mean of ZS@1, TR@1, IR@1).

Rank	Loss	Opt.	ZS@1	TR@1	IR@1	Avg@1
1	iSogCLR	AdamP	27.59	16.10	11.81	**18.50**
2	iSogCLR	RAdam	27.42	16.10	11.70	18.41
3	iSogCLR+TempNet	AdamP	28.69	14.50	11.07	18.09
4	iSogCLR+CyCLIP	SGDP	29.28	14.96	9.63	17.96
5	iSogCLR+CyCLIP	AdamP	27.84	15.04	10.86	17.91
6	iSogCLR	LAMB	26.06	15.82	11.79	17.89
7	iSogCLR	SGDP	28.05	14.26	10.32	17.54
8	iSogCLR+CyCLIP	RAdam	27.00	14.68	10.75	17.48
9	iSogCLR	AdamW	26.48	14.60	10.88	17.32
10	SogCLR	AdamP	25.73	14.76	11.24	17.24

4.5 Key Findings

4.5.1 Global Contrastive Learning Excels

iSogCLR achieves the best overall performance (18.50% Avg@1), outperforming CLIP by 21.4%. SogCLR ranks second among base methods (17.24% Avg@1, +13.1% vs CLIP). This validates that global objectives with moving averages overcome batch-size limitations effectively.

4.5.2 Optimizer Performance

Strong performers across losses: AdamP and RAdam consistently rank in the top-3 across most loss functions (iSogCLR, SogCLR, CyCLIP, SigLIP). AdamP achieves best results for iSogCLR (18.50%), SogCLR (17.24%), and CyCLIP (17.22%). RAdam excels for SigLIP (16.08%) and CLIP (15.24%).

LAMB: Competitive performance with iSogCLR (17.89%, rank 6) and strong retrieval scores across methods.

SGDP: Achieves highest zero-shot accuracy with iSogCLR+CyCLIP (29.28%) but lower retrieval performance.

4.5.3 VICReg Failure

VICReg catastrophically fails across all optimizers (best: 3.49% with AdamW, worst: 0.04% with SGDP and Adafactor), averaging 96.8% below CLIP. The non-contrastive formulation fails to provide sufficient signal for cross-modal alignment without explicit negative sampling. Table 4 shows VICReg's consistent failure.

4.6 Impact of CyCLIP Regularization

We analyze whether adding CyCLIP's geometric constraints (λ_2 for intra-modal, λ_3 for cross-modal consistency) improves performance.

4.6.1 CyCLIP Hyperparameter Selection

We tested multiple (λ_2, λ_3) combinations:

Table 3: Best performance for each loss function with corresponding optimizer.

Loss	Best Opt.	Avg@1	ZS@1	TR@1	IR@1
iSogCLR	AdamP	**18.50**	27.59	16.10	11.81
iSogCLR+TempNet	AdamP	18.09	28.69	14.50	11.07
iSogCLR+CyCLIP	SGDP	17.96	29.28	14.96	9.63
SogCLR	AdamP	17.24	25.73	14.76	11.24
CyCLIP	AdamP	17.22	26.52	14.22	10.93
SigLIP+CyCLIP	RAdam	16.18	24.63	13.90	10.00
SigLIP	RAdam	16.08	24.72	13.82	9.71
CLIP	RAdam	15.24	22.50	13.38	9.82
VICReg	AdamW	3.49	5.96	2.60	1.91

Table 4: VICReg performance across all optimizers shows consistent failure.

Optimizer	ZS@1	TR@1	IR@1
AdamW	5.96	2.60	1.91
AdamP	6.79	3.10	2.44
RAdam	6.67	3.24	2.49
LAMB	5.83	2.92	2.23
Lion	5.23	2.36	1.98
SGDP	0.09	0.02	0.02
Adafactor	0.10	0.00	0.02

- **iSogCLR+CyCLIP**: Best performance at $(0.1, 0.1)$
- **SigLIP+CyCLIP**: Best performance at $(0.5, 0.5)$

4.6.2 Effect on iSogCLR: Zero-Shot Classification Gains

Adding CyCLIP to iSogCLR consistently improves zero-shot ImageNet classification across ALL top-k metrics for most optimizers. Table 5 shows comprehensive zero-shot results:

Table 5: CyCLIP's effect on iSogCLR zero-shot classification across all Top-k metrics ($\lambda_2 = 0.1, \lambda_3 = 0.1$).

Optimizer	iSogCLR				iSogCLR+CyCLIP			
	ZS@1	ZS@3	ZS@5	ZS@10	ZS@1	ZS@3	ZS@5	ZS@10
SGDP	28.05	41.23	46.97	53.93	**29.28**	**43.55**	**49.92**	**57.14**
					+1.23	+2.32	+2.95	+3.21
AdamP	27.59	41.42	47.19	54.89	27.84	43.25	49.90	58.05
					+0.25	+1.83	+2.71	+3.16
AdamW	26.48	39.30	45.13	52.19	26.86	42.22	48.15	55.98
					+0.38	+2.92	+3.02	+3.79
RAdam	27.42	41.42	47.41	54.94	27.00	42.60	49.31	57.75
					-0.42	+1.18	+1.90	+2.81
LAMB	26.06	40.62	47.01	55.22	24.95	40.82	47.84	56.51
					-1.11	+0.20	+0.83	+1.29
Lion	24.83	36.86	42.04	48.76	17.61	30.11	36.30	44.45
					-7.22	-6.75	-5.74	-4.31

Comprehensive zero-shot improvements: CyCLIP boosts classification accuracy across ALL top-k metrics for most optimizers:

- **SGDP**: Strongest gains at all levels (+1.23% @1, +2.32% @3, +2.95% @5, +3.21% @10), achieving **29.28% ZS@1** and **57.14% ZS@10** - the highest zero-shot scores across all 56 configurations
- **AdamP**: Consistent improvements (+0.25% @1, +1.83% @3, +2.71% @5, +3.16% @10), with gains increasing at higher k values
- **AdamW**: Strong gains across all metrics (+0.38% @1, +2.92% @3, +3.02% @5, +3.79% @10)
- **RAdam**: Improved at @3, @5, @10 (+1.18% to +2.81%) despite slight @1 decrease (-0.42%)

- **LAMB**: Modest gains at @3, @5, @10 (+0.20% to +1.29%) with @1 decrease (-1.11%)
- **Lion**: Severe degradation across all metrics (-4.31% to -7.22%), indicating incompatibility

Pattern: Improvements are MORE pronounced at higher k values (@5, @10), suggesting CyCLIP's geometric constraints particularly help the model place correct classes within broader top-k predictions, enhancing overall ranking quality beyond just top-1 accuracy.

Table 6 shows the corresponding retrieval performance:

Table 6: CyCLIP's effect on iSogCLR retrieval performance ($\lambda_2 = 0.1, \lambda_3 = 0.1$).

Optimizer	iSogCLR			iSogCLR+CyCLIP		
	TR@1	**IR@1**	**Avg@1**	**TR@1**	**IR@1**	**Avg@1**
SGDP	14.26	10.32	17.54	14.96 +0.70	9.63 -0.69	**17.96** +0.42
AdamP	16.10	11.81	**18.50**	15.04 -1.06	10.86 -0.95	17.91 -0.59
AdamW	14.60	10.88	17.32	12.94 -1.66	10.29 -0.59	16.70 -0.62
RAdam	16.10	11.70	18.41	14.68 -1.42	10.75 -0.95	17.48 -0.93
LAMB	15.82	11.79	17.89	13.78 -2.04	10.43 -1.36	16.39 -1.50
Lion	13.54	10.04	16.14	9.48 -4.06	7.08 -2.96	11.39 -4.75

Retrieval trade-off: Despite zero-shot gains, retrieval performance degrades for most optimizers. The exception is SGDP, which achieves both zero-shot improvement AND slight retrieval gain (+0.70% TR@1), making it the only optimizer where CyCLIP improves overall Avg@1 (+0.42%, 17.54→17.96).

Key insight: The geometric consistency constraints enhance discriminative features crucial for classification (especially at top-5 and top-10 levels) but smooth the similarity space in ways that reduce fine-grained retrieval precision. This suggests different optimization objectives for classification vs retrieval tasks.

4.6.3 Effect on SigLIP: No Consistent Pattern

Adding CyCLIP to SigLIP shows no consistent pattern (Table 7):

Table 7: Effect of adding CyCLIP to SigLIP with $\lambda_2 = 0.5, \lambda_3 = 0.5$.

Optimizer	SigLIP			SigLIP+CyCLIP		
	ZS@1	**TR@1**	**IR@1**	**ZS@1**	**TR@1**	**IR@1**
AdamP	24.69	13.34	9.68	24.85 (+0.16)	13.28 (-0.06)	9.74 (+0.06)
RAdam	24.72	13.82	9.71	24.63 (-0.09)	13.90 (+0.08)	10.00 (+0.29)
AdamW	23.93	12.94	9.21	24.16 (+0.23)	12.68 (-0.26)	9.39 (+0.18)
LAMB	21.86	12.86	9.86	21.05 (-0.81)	12.68 (-0.18)	9.82 (-0.04)

Changes are small and inconsistent across optimizers, suggesting SigLIP's pairwise sigmoid formulation may not benefit substantially from cycle consistency constraints.

4.6.4 Summary: When Does CyCLIP Help

CyCLIP regularization is most effective for:

- **Zero-shot classification tasks**: Consistent improvements when paired with global contrastive methods (iSogCLR) and projection-based optimizers (SGDP, AdamP)
- **SGDP optimizer**: The only case showing improvement in both zero-shot (+1.23%) and retrieval (+0.70% TR@1), achieving 29.28% zero-shot accuracy

- **Distribution shift scenarios**: The zero-shot gains suggest CyCLIP's geometric constraints improve generalization to unseen datasets

CyCLIP is less beneficial when:

- Fine-grained retrieval is the primary goal (consistent TR@1, IR@1 degradation)
- Paired with sign-based optimizers like Lion (severe performance collapse)

4.7 TempNet: Neural Temperature Learning

We additionally evaluated iSogCLR+TempNet with AdamP, which extends iSogCLR by parameterizing temperatures as neural network outputs rather than per-sample scalars. Results (Table 8):

Table 8: iSogCLR+TempNet performance compared to standard iSogCLR.

Method	ZS@1	TR@1	IR@1	Avg@1
iSogCLR+AdamP	27.59	16.10	11.81	18.50
iSogCLR+TempNet+AdamP	**28.69**	14.50	11.07	18.09
Change	+1.10	-1.60	-0.74	-0.41

Zero-shot improvement: TempNet achieves 28.69% zero-shot accuracy (+1.10% vs standard iSogCLR), the highest among all iSogCLR variants with AdamP. This suggests learned temperature functions better generalize to ImageNet's distribution shift.

Retrieval trade-off: Similar to CyCLIP, retrieval performance degrades (TR@1: -1.60%, IR@1: -0.74%), indicating neural temperature learning may over-regularize fine-grained similarities.

Overall ranking: TempNet ranks 3rd overall (18.09% Avg@1), demonstrating neural temperatures are competitive despite added complexity.

5 Discussion

Our systematic comparison of 8 loss functions and 7 optimizers across 56 configurations reveals several key insights about vision-language contrastive learning at moderate scale.

5.1 Why Global Contrastive Learning Succeeds

iSogCLR and SogCLR substantially outperform batch-based methods (CLIP, CyCLIP) by addressing fundamental batch-size limitations. The additional per-sample temperature individualization in iSogCLR (via DRO) allows adaptive focus on hard samples, explaining its superior performance over standard SogCLR.

5.2 The Classification-Retrieval Trade-off

Both CyCLIP regularization and TempNet exhibit a consistent pattern: improved zero-shot classification (+0.25% to +1.23%) at the cost of retrieval performance (-0.74% to -2.04%). This suggests these methods enhance discriminative features beneficial for coarse-grained categorization while smoothing the similarity space in ways that reduce fine-grained matching precision. The trade-off indicates these techniques optimize different aspects of the learned representation.

5.3 VICReg's Cross-Modal Failure

VICReg's catastrophic failure (averaging 3.49% vs 15.24% for CLIP) across all optimizers confirms that non-contrastive objectives designed for unimodal SSL do not transfer to cross-modal scenarios. Without explicit negative sampling or contrastive pressure, the variance-invariance-covariance regularization fails to align image and text spaces meaningfully. This validates the necessity of contrastive or pairwise objectives for vision-language learning.

5.4 Optimizer-Loss Interactions

AdamP and RAdam emerge as reliable choices across diverse loss functions, performing well with both global contrastive methods (iSogCLR, SogCLR) and batch-based approaches (CLIP, SigLIP, CyCLIP). LAMB shows strong performance with iSogCLR, while SGDP achieves the highest zero-shot accuracy when paired with iSogCLR+CyCLIP, suggesting projection-based methods may have specific advantages for certain loss-optimizer combinations.

5.5 Limitations

Hyperparameter tuning: We used scale-adapted hyperparameters from literature rather than per-optimizer grid search. While this provides fair comparison under realistic constraints, individual optimizers (particularly Lion and SGDP) might benefit from extensive tuning.

Training scale: Our 100k subset, 30-epoch, batch-128 setting represents moderate-scale training. Findings may differ at larger scales (millions of samples, hundreds of epochs).

Architecture constraints: Fixed pretrained encoders prevent exploring architecture-optimizer-loss interactions that might emerge with joint training.

6 Conclusion and Future Work

We conducted a systematic comparison of 8 loss functions, 7 optimizers, and 2 hybrid approaches for vision-language contrastive learning, totaling 56 configurations. Our findings establish that global contrastive learning with adaptive per-sample temperature scaling (iSogCLR) paired with adaptive gradient clipping (AdamP) achieves optimal performance (18.50% Avg@1, 27.59% zero-shot ImageNet), substantially outperforming batch-based CLIP. Neural temperature learning (TempNet) achieves the highest zero-shot accuracy (28.69%) among iSogCLR variants, though with retrieval trade-offs.

Key contributions include: (1) demonstrating global contrastive methods' superiority over batch-based approaches at moderate scale, (2) revealing consistent classification-retrieval trade-offs when adding geometric constraints or neural temperature learning, (3) confirming VICReg's failure for cross-modal alignment across all optimizers, and (4) identifying AdamP and RAdam as robust optimizer choices across diverse loss functions.

6.1 Future Directions

Curriculum learning strategies:

- **Similarity-based curriculum**: Use pretrained models to score image-text pair similarity, training on easy (high-similarity) pairs before hard (low-similarity) pairs
- **TempNet-driven curriculum**: Leverage TempNet's learned temperatures as dynamic difficulty signals for curriculum sampling, loss weighting, or adaptive augmentation strength

Reinforcement learning for adaptive training:

- **RL for sample selection**: Formulate data sampling as a decision problem, using Q-learning to choose samples (easy/hard/noisy/rare) that maximize learning progress
- **RL for temperature control**: Train TempNet using policy-based or Q-learning methods to adaptively select temperatures, optimizing for training stability and convergence speed

Scaling and architecture exploration: Evaluating these methods at larger scales (millions of samples, larger batch sizes) and with trainable encoders could reveal additional insights into loss-optimizer-architecture interactions.

Hybrid losses: Experiment with more contrastive and non-contrastive hybrid losses.

7 Contributions

This project was completed collaboratively by two team members with equal contribution to implementation, experimentation, and presentation. Varsha was responsible for all code changes for new losses. Ran experiments on iSogCLR, iSogCLR with Tempnet and hybrid losses. Kanishk conducted experiments on SogCLR, CLIP, CyCLIP, VicREG. The final report and presentation were written jointly. For the demo, please visit the following link: Demo Video.

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A Complete Experimental Results

This appendix provides comprehensive results for all experimental configurations, enabling full reproducibility and detailed analysis.

A.1 Zero-Shot ImageNet Classification Results

Table 9 presents zero-shot classification accuracy on ImageNet validation set, sorted by ZS@1.

Table 9: Complete zero-shot ImageNet classification results sorted by Top-1 accuracy.

Rank	Loss	Optimizer	ZS@1	ZS@3	ZS@5	ZS@10
1	iSogCLR+CyCLIP	SGDP	29.284	43.548	49.916	57.14
2	iSogCLR+TempNet	AdamP	28.694	43.352	49.42	56.644
3	iSogCLR	SGDP	28.05	41.228	46.97	53.93
4	iSogCLR+CyCLIP	AdamP	27.836	43.248	49.896	58.05
5	iSogCLR	AdamP	27.586	41.416	47.192	54.894
6	iSogCLR	RAdam	27.422	41.416	47.414	54.936
7	iSogCLR+CyCLIP	RAdam	27.0	42.598	49.308	57.75
8	iSogCLR+CyCLIP	AdamW	26.862	42.222	48.146	55.98
9	CyCLIP	AdamP	26.518	40.816	47.524	55.514
10	iSogCLR	AdamW	26.482	39.298	45.13	52.192
11	CyCLIP	RAdam	26.466	40.648	47.142	55.608
12	SogCLR	SGDP	26.604	39.612	45.436	52.806
13	iSogCLR	LAMB	26.056	40.624	47.006	55.216
14	SogCLR	AdamP	25.73	39.524	45.658	53.5
15	CyCLIP	AdamW	25.728	39.294	45.352	53.308
16	SogCLR	RAdam	25.576	39.228	45.288	53.136
17	iSogCLR+CyCLIP	LAMB	24.95	40.818	47.836	56.51
18	SigLIP+CyCLIP	AdamP	24.85	38.538	44.656	52.714
19	iSogCLR	Lion	24.834	36.858	42.036	48.764
20	SigLIP	RAdam	24.724	38.63	44.872	52.6
21	SogCLR	AdamW	24.698	37.822	43.408	50.916
22	SigLIP	AdamP	24.686	38.416	44.572	52.646
23	SigLIP+CyCLIP	RAdam	24.632	38.498	44.826	53.162
24	SigLIP+CyCLIP	AdamW	24.162	37.028	42.574	49.996
25	SigLIP	AdamW	23.93	37.102	42.426	49.82
26	SogCLR	LAMB	23.686	37.516	44.102	52.646
27	CyCLIP	LAMB	23.656	38.036	45.056	53.854
28	CyCLIP	Lion	22.946	35.956	41.52	49.192
29	CLIP	RAdam	22.502	35.942	42.352	51.166
30	CLIP	AdamP	22.61	35.982	42.182	50.814
31	SogCLR	Lion	22.37	34.36	39.646	46.81
32	CLIP	AdamW	21.906	34.846	40.946	48.948
33	SigLIP	LAMB	21.858	36.052	42.8	51.04
34	SigLIP+CyCLIP	LAMB	21.052	35.396	42.158	50.784
35	CLIP	LAMB	20.478	34.028	40.754	49.866
36	CLIP	Lion	20.408	32.498	37.946	45.356
37	SigLIP+CyCLIP	Lion	20.298	32.654	38.142	45.468
38	SigLIP	Lion	20.27	32.538	37.908	45.436
39	iSogCLR+CyCLIP	Lion	17.612	30.106	36.298	44.45
40	SogCLR	Adafactor	14.07	24.426	29.428	36.524
41	CLIP	Adafactor	13.15	23.204	28.414	35.678
42	iSogCLR	AdamW (bad LR)	12.83	22.298	26.982	33.586
43	VICReg	AdamP	6.792	15.038	20.654	30.052
44	VICReg	RAdam	6.674	15.16	20.542	30.36
45	VICReg	AdamW	5.96	12.902	17.774	26.76
46	VICReg	LAMB	5.828	13.102	18.424	27.498
47	VICReg	Lion	5.228	11.92	16.346	24.898
48	CLIP	SGDP	2.45	6.094	8.784	13.724
49	VICReg	Adafactor	0.1	0.308	0.506	1.03
50	VICReg	SGDP	0.092	0.272	0.456	0.97

A.2 Text-to-Image Retrieval Results

Table 10 presents text-to-image retrieval performance on MSCOCO validation set, sorted by TR@1.

A.3 Image-to-Text Retrieval Results

Table 11 presents image-to-text retrieval performance on MSCOCO validation set, sorted by IR@1.

A.4 Summary Statistics

Table 12 presents overall performance ranked by Average Recall@1.

Table 10: Complete text-to-image retrieval results sorted by Recall@1.

Rank	Loss	Optimizer	TR@1	TR@5	TR@10
1	iSogCLR	AdamP	16.1	37.34	48.92
2	iSogCLR	RAdam	16.1	37.78	49.16
3	iSogCLR	LAMB	15.82	36.76	48.98
4	iSogCLR+CyCLIP	AdamP	15.04	34.44	46.1
5	iSogCLR+CyCLIP	SGDP	14.96	35.04	46.42
6	SogCLR	LAMB	14.82	35.5	47.5
7	SogCLR	RAdam	14.8	35.42	47.84
8	SogCLR	AdamP	14.76	35.04	47.66
9	iSogCLR+CyCLIP	RAdam	14.68	35.16	45.56
10	iSogCLR	AdamW	14.6	34.24	45.98
11	iSogCLR+TempNet	AdamP	14.5	34.86	46.4
12	CyCLIP	LAMB	14.48	35.2	47.46
13	iSogCLR	SGDP	14.26	35.54	47.12
14	CyCLIP	AdamP	14.22	34.42	47.58
15	CyCLIP	RAdam	14.12	34.78	47.08
16	SigLIP+CyCLIP	RAdam	13.9	34.52	47.24
17	SigLIP	RAdam	13.82	34.38	46.3
18	iSogCLR+CyCLIP	LAMB	13.78	33.82	45.52
19	CyCLIP	AdamW	13.72	34.4	46.44
20	iSogCLR	Lion	13.54	31.9	43.8
21	SogCLR	SGDP	13.52	34.04	45.5
22	CLIP	RAdam	13.38	32.86	44.62
23	SigLIP	AdamP	13.34	34.72	47.36
24	SigLIP+CyCLIP	AdamP	13.28	34.68	46.42
25	CLIP	LAMB	13.24	33.62	46.14
26	SogCLR	AdamW	13.08	32.82	44.18
27	SigLIP	AdamW	12.94	32.52	44.34
28	iSogCLR+CyCLIP	AdamW	12.94	32.54	44.1
29	SigLIP	LAMB	12.86	33.18	45.88
30	CLIP	AdamP	12.72	33.22	44.88
31	SigLIP+CyCLIP	LAMB	12.68	33.44	45.02
32	SigLIP+CyCLIP	AdamW	12.68	32.18	43.74
33	CLIP	AdamW	12.22	31.98	43.78
34	CyCLIP	Lion	12.08	30.96	42.74
35	SogCLR	Lion	11.24	29.9	41.68
36	SigLIP	Lion	11.04	29.72	41.18
37	SigLIP+CyCLIP	Lion	10.66	28.44	40.26
38	CLIP	Lion	10.56	27.42	39.36
39	iSogCLR+CyCLIP	Lion	9.48	26.32	37.5
40	SogCLR	Adafactor	6.36	18.12	26.86
41	CLIP	Adafactor	6.36	18.96	27.7
42	iSogCLR	AdamW (bad LR)	5.64	17.56	26.38
43	VICReg	RAdam	3.24	10.52	17.1
44	VICReg	AdamP	3.1	11.08	16.88
45	VICReg	LAMB	2.92	9.7	15.14
46	VICReg	AdamW	2.6	8.86	14.26
47	VICReg	Lion	2.36	8.98	14.06
48	CLIP	SGDP	1.2	4.44	7.62

Table 11: Complete image-to-text retrieval results sorted by Recall@1.

Rank	Loss	Optimizer	IR@1	IR@5	IR@10
1	iSogCLR	AdamP	11.807	29.413	40.074
2	iSogCLR	LAMB	11.788	29.541	41.033
3	iSogCLR	RAdam	11.704	29.593	40.721
4	SogCLR	LAMB	11.368	29.114	40.170
5	SogCLR	AdamP	11.24	28.674	39.806
6	CyCLIP	LAMB	11.128	28.682	39.562
7	iSogCLR+TempNet	AdamP	11.068	27.690	38.782
8	SogCLR	RAdam	11.008	28.490	39.486
9	CyCLIP	RAdam	10.952	28.358	39.454
10	CyCLIP	AdamP	10.932	28.082	39.206
11	iSogCLR	AdamW	10.880	27.762	38.366
12	iSogCLR+CyCLIP	AdamP	10.864	27.358	38.346
13	iSogCLR+CyCLIP	RAdam	10.748	27.578	38.594
14	iSogCLR+CyCLIP	LAMB	10.428	27.026	38.178
15	CyCLIP	AdamW	10.360	27.274	38.114
16	iSogCLR	SGDP	10.320	26.718	37.323
17	iSogCLR+CyCLIP	AdamW	10.288	26.047	36.367
18	SogCLR	AdamW	10.268	26.670	37.670
19	SogCLR	SGDP	10.236	26.786	37.690
20	iSogCLR	Lion	10.036	25.855	36.279
21	SigLIP+CyCLIP	RAdam	10.004	26.862	37.878
22	CLIP	AdamP	9.972	26.482	37.211
23	CLIP	LAMB	9.872	26.902	38.102
24	SigLIP	LAMB	9.860	26.554	37.487
25	CLIP	RAdam	9.824	26.786	37.826
26	SigLIP+CyCLIP	LAMB	9.820	26.482	37.906
27	SigLIP+CyCLIP	AdamP	9.736	26.422	37.786
28	SigLIP	RAdam	9.709	26.323	37.403
29	SigLIP	AdamP	9.677	26.119	37.459
30	iSogCLR+CyCLIP	SGDP	9.628	25.322	35.535
31	CLIP	AdamW	9.405	25.203	35.867
32	SigLIP+CyCLIP	AdamW	9.385	25.515	36.359
33	SogCLR	Lion	9.357	24.439	34.652
34	SigLIP	AdamW	9.209	25.095	36.079
35	CyCLIP	Lion	9.057	24.455	34.751
36	SigLIP	Lion	8.233	23.332	33.496
37	CLIP	Lion	8.209	22.676	33.024
38	SigLIP+CyCLIP	Lion	8.149	22.648	32.804
39	iSogCLR+CyCLIP	Lion	7.077	20.652	30.4
40	CLIP	Adafactor	5.230	15.902	24.035
41	SogCLR	Adafactor	5.214	15.986	24.231
42	iSogCLR	AdamW (bad LR)	4.934	15.247	23.388
43	VICReg	RAdam	2.491	8.201	13.303
44	VICReg	AdamP	2.443	8.153	13.471
45	VICReg	LAMB	2.227	7.473	12.260
46	VICReg	Lion	1.975	6.790	11.244
47	VICReg	AdamW	1.911	6.614	11.204
48	CLIP	SGDP	0.948	4.222	7.533

Table 12: Top 25 configurations by Average Recall@1 (mean of ZS@1, TR@1, IR@1).

Rank	**Loss**	**Optimizer**	**ZS@1**	**TR@1**	**IR@1**	**Avg@1**
1	iSogCLR	AdamP	27.586	16.1	11.807	**18.498**
2	iSogCLR	RAdam	27.422	16.1	11.704	18.409
3	iSogCLR+TempNet	AdamP	28.694	14.5	11.068	18.087
4	iSogCLR+CyCLIP	SGDP	29.284	14.96	9.628	17.957
5	iSogCLR+CyCLIP	AdamP	27.836	15.04	10.864	17.913
6	iSogCLR	LAMB	26.056	15.82	11.788	17.888
7	iSogCLR	SGDP	28.05	14.26	10.320	17.543
8	iSogCLR+CyCLIP	RAdam	27.0	14.68	10.748	17.476
9	iSogCLR	AdamW	26.482	14.6	10.880	17.321
10	SogCLR	AdamP	25.73	14.76	11.24	17.243
11	CyCLIP	AdamP	26.518	14.22	10.932	17.223
12	CyCLIP	RAdam	26.466	14.12	10.952	17.179
13	SogCLR	RAdam	25.576	14.8	11.008	17.128
14	SogCLR	SGDP	26.604	13.52	10.236	16.787
15	iSogCLR+CyCLIP	AdamW	26.862	12.94	10.288	16.697
16	SogCLR	LAMB	23.686	14.82	11.368	16.625
17	CyCLIP	AdamW	25.728	13.72	10.360	16.603
18	CyCLIP	LAMB	23.656	14.48	11.128	16.421
19	iSogCLR+CyCLIP	LAMB	24.95	13.78	10.428	16.386
20	iSogCLR	Lion	24.834	13.54	10.036	16.137
21	SogCLR	AdamW	24.698	13.08	10.268	16.015
22	SigLIP+CyCLIP	RAdam	24.632	13.9	10.004	16.179
23	SigLIP	RAdam	24.724	13.82	9.709	16.084
24	SigLIP+CyCLIP	AdamP	24.85	13.28	9.736	15.955
25	SigLIP	AdamW	23.93	12.94	9.209	15.360